

Xiaoyu Zhang

CONTACT INFORMATION

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RESEARCH SUMMARY

I build **multimodal sensing agents** that *perceive, reason, and act* on the physical world by jointly leveraging sensing modalities organized across four complementary physical domains: **radio-frequency** (mmWave FMCW radar, synthetic-aperture and cascaded-array imaging), **acoustic / ultrasonic** (photoacoustic imaging, ultrasound, audio and speech), **optical / radiographic** (visible-light vision, infrared, mammography), and **bio-electrical** (wearable ear-EEG). My research closes the loop from **novel sensing front-ends** through **physics-informed inverse imaging** and **cross-modal representation learning** to **deployed clinical systems**, with validated impact on **smart health** (wound care, breast cancer subtyping, thyroid diagnosis, Alzheimer’s screening, hearing disorders) and **social good** (sustainable building, accessible HCI, point-of-care diagnostics). My work has produced **15 peer-reviewed papers** (7 first-authored, at venues including ACM IMWUT, ACM UIST, IEEE IoTJ, IEEE/RSJ IROS, and npj Imaging), a **Best Paper Candidate** at ACM SenSys HumanSys’25, **4 patents** (one under commercialization by a medical-device startup, whose technology I pitched to the Panasci TEC 2025 semifinals), and a **clinical station deployed under an IRB-approved validation study**, reflecting a deliberate *research–clinic–industry* translation pipeline.

RESEARCH INTERESTS

My research lies at the intersection of **Wireless Sensing, Multimodal Medical AI, and Embodied Intelligence**. One technical question recurs across all three: *when the sensor is cheap and the data are scarce, where should the burden of perception sit — on physics, on learning, or on the hardware aperture?*

1) mmWave Sensing and Imaging. Turning low-cost commodity radar into a high-resolution instrument for sensing the human body, by shifting the burden of resolution onto embedded physics, learned cross-modal supervision, or a synthesized aperture.

- *Clinical imaging* — wound assessment through dressings; tissue moisture and perfusion mapping
- *Human–machine interaction* — hand and gesture tracking for touchless control in sterile or low-light settings
- *Identity and affect* — mmWave–voice multi-factor identification; micro-expression sensing
- *Privacy-preserving monitoring* — presence and activity sensing that carries no identifying imagery

2) Multimodal Medical AI. Fusing and transferring across imaging and physiological signals where labeled clinical data are scarce, and testing whether a pretrained foundation model *genuinely* transfers to a new clinical population.

- *Oncology* — non-invasive breast-cancer molecular subtyping from photoacoustic, ultrasound and mammographic imaging
- *Endocrine* — unified thyroid nodule diagnosis and lesion segmentation from clinical ultrasound
- *Wound and vascular* — chronic foot-ulcer prognosis and perfusion prediction from photoacoustic imaging
- *Neurological and auditory* — Alzheimer’s subtype screening from speech; sound-sensitivity screening from wearable ear-EEG

3) Robotic Perception and Autonomous Inspection. Equipping autonomous platforms to assess material and structural condition non-destructively, and to close a control loop on that assessment.

- *Building energy* — autonomous robotic inspection of thermal insulation beneath building surfaces

- *Sustainable manufacturing* — closed-loop visual process control for construction-material production

EDUCATION

University at Buffalo, the State University of New York (SUNY) Sep. 2021 - Jul. 2026
 Ph.D., Computer Science and Engineering Co-advised by Prof. Wen Yao Xu & Yaxiong Xie
 Dissertation (**defended Jul. 2026**): *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications*
 Committee: Wen Yao Xu, Yaxiong Xie, Jun Xia

University of Science and Technology of China Sep. 2017 - Jun. 2020
 M.S., Electronics and Communication Engineering Advised by Prof. Bin Liu

Hefei University of Technology Sep. 2013 - Sep. 2017
 B.Eng., Electronic Information Engineering

PUBLICATIONS

Summary. 15 peer-reviewed papers, 7 first-authored (name in **bold**), spanning flagship venues in ubiquitous computing (ACM IMWUT, UIST), robotics (IEEE/RSJ IROS), medical imaging (Photoacoustics, npj Imaging), and biomedical informatics (IEEE JBHI, IoTJ). Thrust tags: [\[Health\]](#) [\[Embodied\]](#) [\[Social Good\]](#).

- [IROS'26] [\[Social Good\]](#) **Xiaoyu Zhang**, Ye Zhan, Chenhan Xu, Zhengxiong Li, Chuqin Huang, Yanda Cheng, Licheng Liang, Wei Bo, Jun Xia, Chi Zhou, Hongyue Sun, Wen Yao Xu, “*Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing*”, IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. (*Patent filed.*)
- [IMWUT'26] [\[Health\]](#) Shuwei Hou, Wei Bo, L. Cao, Varun Shijo, Chuhui Liu, **Xiaoyu Zhang**, L. Guo, Wen Yao Xu, “*Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis*”, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2026.
- [ICIBM'26] [\[Health\]](#) **Xiaoyu Zhang**, Wei Bo, Anarghya Das, Longxiang Pan, Wen Yao Xu, “*Matching Speech Models to AD/HD Subtypes: A Mechanistic Benchmark*”, International Conference on Intelligent Biology and Medicine, 2026.
- [npj Imaging'26] [\[Health\]](#) Yanda Cheng, Chuqin Huang, Shuliang Yu, Saptarshi Chakraborty, Yunqi Xi, Robert W. Bing, Huijuan Zhang, **Xiaoyu Zhang**, Isabel Komornicki, Linda M. Harris, Wen Yao Xu, Jun Xia, “*Predictive modeling of chronic foot ulcer outcomes using longitudinal photoacoustic imaging*”, npj Imaging, 2026.
- [HealthCom'26] [\[Health\]](#) Nicholas Wei, Austin Xu, **Xiaoyu Zhang**, “*Pool or Specialize? Anatomic-Site Domain Shift in Melanoma Classification*”, IEEE International Conference on E-health Networking, Application & Services, 2026.
- [IoTJ'25] [\[Health\]](#) **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wen Yao Xu, “*mmSkin: An Over-gauze Wound Assessment System using Radio Frequency Technologies*”, IEEE Internet of Things Journal, 2025. (*Translated into US patent; under commercialization by a medical-device startup.*)
- [IoTJ'25] [\[Embodied\]](#) **Xiaoyu Zhang**, Zhengxiong Li, Chenhan Xu, Luchuan Song, Huining Li, Hongfei Xue, Yingxiao Wu, Wen Yao Xu, “*mmHand: Towards Pixel-Level-Accuracy Hand Localization Using a Single Commodity mmWave Device*”, IEEE Internet of Things Journal, 2025.
- [HumanSys'25] [\[Health\]](#) **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wen Yao Xu, “*Through-dressing Wound Monitoring Based on*

the mmWave Sensor”, The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM SenSys), 2025. (**Best Paper Candidate**)

- [Sensors’25] [Health] **Xiaoyu Zhang**, Chuhui Liu, Yanda Cheng, Zhengxiong Li, Chenhan Xu, Chuqin Huang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “A Comprehensive Survey of Research Trends in mmWave Technologies for Medical Applications”, Sensors, 2025.
- [Photoacoustics’25] [Health] Chuqin Huang, Yanda Cheng, **Xiaoyu Zhang**, Ye Zhan, Wenhan Zheng, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, “Radiomics-Driven Perfusion Prediction in Clinical Photoacoustic Foot Imaging”, Photoacoustics, 2025.
- [Photoacoustics’25] [Health] Chuqin Huang, Emily Zheng, Wenhan Zheng, Huijuan Zhang, Yanda Cheng, **Xiaoyu Zhang**, Varun Shijo, Robert W. Bing, Isabel Komornicki, Linda M. Harris, Ermelinda Bonaccio, Kazuaki Takabe, Emma Zhang, Wenyao Xu, Jun Xia, “Enhanced clinical photoacoustic vascular imaging through a skin localization network and adaptive weighting”, Photoacoustics, 2025.
- [JBHI’24] [Health] Wei Bo, Suzanne S. Sullivan, **Xiaoyu Zhang**, Mingchen Gao, Wenyao Xu, “A Telemedicine Analytic Framework for Fully and Semi-automatic Alzheimer’s Disease Screening using Clock Drawing Test”, IEEE Journal of Biomedical and Health Informatics, 2024.
- [UIST’23] [Embodied] Tiantian Liu, Feng Lin, Chao Wang, Chenhan Xu, **Xiaoyu Zhang**, Zhengxiong Li, Wenyao Xu, Ming-Chun Huang, Kui Ren, “WaveID: Robust and Secure Multi-modal User Identification via mmWave-voice Mechanism”, ACM Symposium on User Interface Software and Technology, San Francisco, USA, 2023.
- [BSN’19] [Embodied] **Xiaoyu Zhang**, Bin Liu, “A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs”, IEEE-EMBS International Conference on Body Sensor Networks, Chicago, USA, 2019.
- [Bodynets’17] [Embodied] Chengjie Guan, Bin Liu, Zhiqiang Liu, Y. Zhang, **Xiaoyu Zhang**, “JMMM: A Mobility Model for WBANs Based on Human Joint Movements”, 19th EAI International Conference on Body Area Networks, Dalian, China, 2017.

HONORS AND AWARDS

- **Best Paper Candidate Award**, HumanSys’25 — The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (co-located with ACM SenSys), 2025
- **Semifinalist**, Panasci Technology Entrepreneurship Competition (Panasci TEC), University at Buffalo, 2025
- **Graduate Teaching Award**, Department of CSE, University at Buffalo, 2024
- **Chair’s Fellowship**, Department of CSE, University at Buffalo, 2021
- Merit scholarships (graduate and undergraduate), USTC and HFUT, 2015–2019

INDUSTRY & CLINICAL EXPERIENCE

Clinical Research Collaborations

2022 - Present

Multi-year IRB-approved clinical collaborations at the University at Buffalo and partner sites, spanning device deployment, on-site data collection, and multi-modal cohort curation:

- **Wound care** (UBMD Vascular Surgery) — deployed the WaveSight-RF clinical station on site; **5 patients** scanned to date under an IRB-approved protocol, with repeat longitudinal scanning up to **3 months** per patient
- **Breast imaging** — clinical cohort of **284 imaged patients**; curated the **53-patient** tri-modal photoacoustic / ultrasound / mammography subset used for molecular subtyping
- **Wearable ear-EEG** — **27 subjects** recorded across four passive auditory tasks (~1,800 trials), **23** retained after per-channel quality control, paired with standardized clinical score scales
- **Thyroid ultrasound** — ongoing clinical scanning collaboration for nodule diagnosis and lesion segmentation

- **Photoacoustic Imaging Lab** (Dept. of Biomedical Engineering) — joint imaging-system development and radiomics analysis

China Merchants Bank Software Center

Jul. 2020 - Sep. 2020

Designed and implemented a full-stack data-verification system (Bootstrap + Django) with dynamic front-back interaction for multi-source financial data.

COMPETITION

- **Panasci Technology Entrepreneurship Competition (Panasci TEC) 2025** Mar. 2025
Advanced Through-Gauze Wound Assessment Technology.
Xiaoyu Zhang (technical lead and presenter), Chuhui Liu, Jun Xia, Linda M. Harris, Wenya Xu.
Advanced to the **semifinals**; pitched the patented mmWave wound-assessment platform to clinical and industry judges.

PRESENTATIONS & DEMOS

- **Ph.D. Dissertation Defense:** *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications* Jul. 2026
University at Buffalo, CSE
- **Live Demonstration:** SUNY Research Expo Mar. 2026
WaveSight-RF clinical wound-imaging station — translational outreach
- **Community Engagement Studio:** UB CTSI Feb. 2026
Wound-measurement device design review with a 12-member community expert panel
- **Invited Talk:** The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys, co-located with ACM SenSys) May 2025
Through-dressing Wound Monitoring Based on the mmWave Sensor
- **Pitch Talk:** Panasci Technology Entrepreneurship Competition Mar. 2025
Advanced Through-Gauze Wound Assessment Technology
- **Invited Talk:** IEEE-EMBS International Conference on Body Sensor Networks May 2019
A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs
- **Invited Talk:** 19th EAI International Conference on Body Area Networks Sep. 2017
JMMM: A Mobility Model for WBANs Based on Human Joint Movements

TEACHING EXPERIENCES

Teaching Assistant across **6 semesters** at the University at Buffalo, including one invited guest lecture, plus one semester at USTC.

1. Algorithm Analysis and Design, UB Fall 2021, Spring 2022, Fall 2022, Fall 2023, Fall 2024
2. Special Topics, UB — incl. **invited guest lecture:** *Wireless Signal Processing: Making Sense of the Invisible* Fall 2023
3. Mathematical Logic and Graph Theory, USTC Fall 2018

MENTORING EXPERIENCES

I have mentored **10+ students** on wireless sensing, embedded systems, and AI-based medical screening projects. **Four high-school students I mentored are first authors on peer-reviewed papers.**

- Prakriti Setlur — AI-based clinical image screening pipelines
- Kh Shafin Farhan — foundation models for medical imaging
- Abhi Ramtel (CSE@UB) — UI design of an over-gauze wound assessment system
- Cole Desimone (AE@UB) — 3D model design of the mmWave sensor scanning system
- George Gillman (EE@UB) — mmWave technologies for medical applications: a review
- Weida Jiang (CSE@UB) — embedded mmWave-based hand detection on Raspberry Pi
- Wenxuan Huang & Yiwen Tan (CSE@UB) — tooling for automatic image labeling

COMMUNITY
SERVICES &
OUTREACH
ACTIVITIES

- Nicholas Wei & Austin Xu (high school) — anatomic-site domain shift in melanoma classification
- Henry Yang & Emily Liu (high school) — real-world study of sport-based support for autism spectrum disorder

Conference Organization:

- **Organizing Committee Member (Local Arrangements)**, International Conference on Intelligent Biology and Medicine (ICIBM) [2026]

Journal Reviewer:

- IEEE Transactions on Mobile Computing (TMC) [2025, 2026]
- ACM Transactions on Computing for Healthcare (HEALTH) [2025]
- IEEE Open Journal of Engineering in Medicine and Biology (OJEMB) [2025]
- Scientific Reports [2025]
- Smart Health (SH) [2024, 2025]

Conference Reviewer:

- ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) [2025]
- IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) [2024, 2026]
- Physics Education Research Conference (PERC), American Association of Physics Teachers (AAPT) [2026]
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) [2023, 2024]

Outreach:

- Live technology demonstration, **SUNY Research Expo** — translational outreach for the WaveSight-RF clinical wound-imaging station [Mar. 2026]
- **Community Engagement Studio**, UB Clinical and Translational Science Institute (CTSI) — convened a **12-member community expert panel** (healthcare providers, clinical trainees, community advocates) for a design review of the wound-measurement device [Feb. 2026]